

Assignment 9: Time Series Analysis

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A09_TimeSeries.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up

1. Set up your session:
 - Check your working directory
 - Load the tidyverse, zoo, and trend packages
 - Set your ggplot theme

```
getwd()
```

```
## [1] "/home/guest/EDE_Spring2026_Shiiyi"
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.2      v tibble     3.2.1
## v lubridate  1.9.3      v tidyr      1.3.1
## v purrr      1.0.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)
library(zoo)
```

```
##
## Attaching package: 'zoo'
##
## The following objects are masked from 'package:base':
##
##      as.Date, as.Date.numeric
```

```
library(trend)
library(here)
```

```
## here() starts at /home/guest/EDE_Spring2026_Shiyi
```

```
mytheme <- theme_classic(base_size = 12) +
  theme(axis.text = element_text(color = "black"),
        legend.position = "top")
theme_set(mytheme)
```

2. Import the ten datasets from the Ozone_TimeSeries folder in the Raw data folder. These contain ozone concentrations at Garinger High School in North Carolina from 2010-2019 (the EPA air database only allows downloads for one year at a time). Import these either individually or in bulk and then combine them into a single dataframe named **GaringerOzone** of 3589 observation and 20 variables.

```
#1
ozone2010 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2010_raw.csv")
ozone2011 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2011_raw.csv")
ozone2012 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2012_raw.csv")
ozone2013 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2013_raw.csv")
ozone2014 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2014_raw.csv")
ozone2015 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2015_raw.csv")
ozone2016 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2016_raw.csv")
ozone2017 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2017_raw.csv")
ozone2018 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2018_raw.csv")
ozone2019 <- read.csv("./Data/Raw/Ozone_TimeSeries/EPAair_03_GaringerNC2019_raw.csv")

GaringerOzone <- bind_rows(ozone2010, ozone2011, ozone2012, ozone2013, ozone2014,
                          ozone2015, ozone2016, ozone2017, ozone2018, ozone2019)

dim(GaringerOzone)
```

```
## [1] 3589 20
```

Wrangle

3. Set your date column as a date class.
4. Wrangle your dataset so that it only contains the columns Date, Daily.Max.8.hour.Ozone.Concentration, and DAILY_AQI_VALUE.

5. Notice there are a few days in each year that are missing ozone concentrations. We want to generate a daily dataset, so we will need to fill in any missing days with NA. Create a new data frame that contains a sequence of dates from 2010-01-01 to 2019-12-31 (hint: `as.data.frame(seq())`). Call this new data frame `Days`. Rename the column name in `Days` to “Date”.
6. Use a `left_join` to combine the data frames. Specify the correct order of data frames within this function so that the final dimensions are 3652 rows and 3 columns. Call your combined data frame `GaringerOzone`.

```
# 3
GaringerOzone$Date <- as.Date(GaringerOzone$Date, format = "%m/%d/%Y")

# 4
GaringerOzone <- GaringerOzone %>%
  select(Date, Daily.Max.8.hour.Ozone.Concentration, DAILY_AQI_VALUE)

# 5
Days <- as.data.frame(seq(as.Date("2010-01-01"),
                          as.Date("2019-12-31"),
                          by = "day"))
names(Days) <- "Date"

# 6
GaringerOzone <- left_join(Days, GaringerOzone, by = "Date")
dim(GaringerOzone)
```

```
## [1] 3652    3
```

Visualize

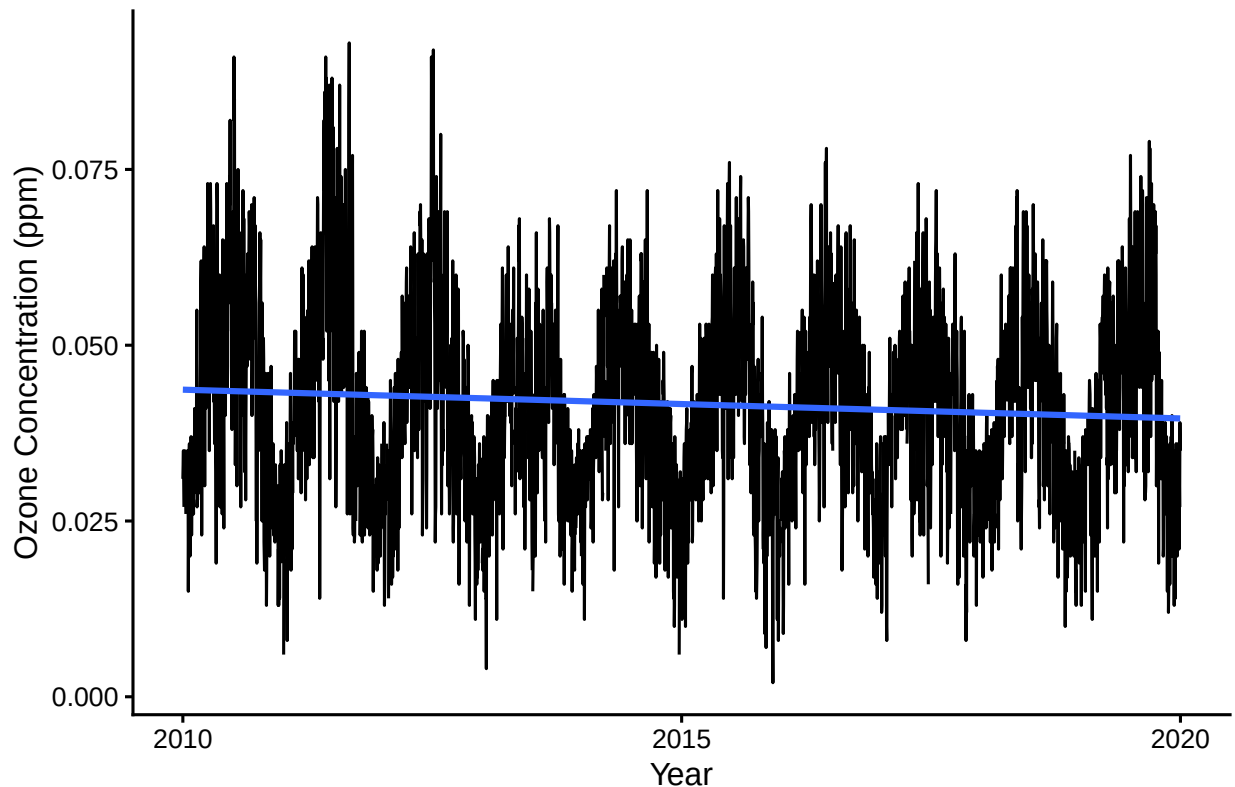
7. Create a line plot depicting ozone concentrations over time. In this case, we will plot actual concentrations in ppm, not AQI values. Format your axes accordingly. Add a smoothed line showing any linear trend of your data. Does your plot suggest a trend in ozone concentration over time?

```
#7
ggplot(GaringerOzone,
       aes(x = Date, y = Daily.Max.8.hour.Ozone.Concentration)) +
  geom_line() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Daily Ozone Concentration at Garinger High School (2010-2019)",
       x = "Year",
       y = "Ozone Concentration (ppm)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
## Warning: Removed 63 rows containing non-finite outside the scale range
## ('stat_smooth()').
```

Daily Ozone Concentration at Garinger High School (2010–2019)



Answer: Yes. The time series plot shows substantial variability in ozone concentrations over time, with clear seasonal fluctuations characterized by recurring peaks and troughs each year. The fitted linear trend line suggests a slight decreasing trend in ozone concentration from 2010 to 2019. However, this downward trend is relatively modest compared to the overall variability and strong seasonal patterns in the data.

Time Series Analysis

Study question: Have ozone concentrations changed over the 2010s at this station?

8. Use a linear interpolation to fill in missing daily data for ozone concentration. Why didn't we use a piecewise constant or spline interpolation?

#8

```
GaringerOzone <- GaringerOzone %>%  
  mutate(Ozone.clean = zoo::na.approx(Daily.Max.8.hour.Ozone.Concentration))
```

Answer: The current result estimates missing values based on adjacent observations and maintains continuity in the time series. Piecewise constant interpolation was not used because it assumes no change between observations, which is unrealistic for ozone concentrations. Spline interpolation was also avoided because it can introduce excessive smoothing and artificial variability, potentially distorting the true pattern of the data.

9. Create a new data frame called `GaringerOzone.monthly` that contains aggregated data: mean ozone concentrations for each month. In your pipe, you will need to first add columns for year and month to form the groupings. In a separate line of code, create a new `Date` column with each month-year combination being set as the first day of the month (this is for graphing purposes only)

```
#9
GaringerOzone.monthly <- GaringerOzone %>%
  mutate(Year = year(Date),
         Month = month(Date)) %>%
  group_by(Year, Month) %>%
  summarise(Ozone.mean = mean(Ozone.clean, na.rm = TRUE)) %>%
  ungroup()
```

```
## 'summarise()' has grouped output by 'Year'. You can override using the
## '.groups' argument.
```

```
GaringerOzone.monthly$Date <- as.Date(paste(GaringerOzone.monthly$Year,
                                           GaringerOzone.monthly$Month,
                                           "01", sep = "-"))
head(GaringerOzone.monthly)
```

```
## # A tibble: 6 x 4
##   Year Month Ozone.mean Date
##   <dbl> <dbl>     <dbl> <date>
## 1  2010     1     0.0305 2010-01-01
## 2  2010     2     0.0345 2010-02-01
## 3  2010     3     0.0446 2010-03-01
## 4  2010     4     0.0556 2010-04-01
## 5  2010     5     0.0466 2010-05-01
## 6  2010     6     0.0576 2010-06-01
```

```
dim(GaringerOzone.monthly)
```

```
## [1] 120  4
```

10. Generate two time series objects. Name the first `GaringerOzone.daily.ts` and base it on the dataframe of daily observations. Name the second `GaringerOzone.monthly.ts` and base it on the monthly average ozone values. Be sure that each specifies the correct start and end dates and the frequency of the time series.

```
#10
GaringerOzone.daily.ts <- ts(GaringerOzone$Ozone.clean,
                             start = c(2010, 1),
                             end = c(2019, 365),
                             frequency = 365)
start(GaringerOzone.daily.ts)
```

```
## [1] 2010  1
```

```
end(GaringerOzone.daily.ts)
```

```
## [1] 2019 365
```

```
frequency(GaringerOzone.daily.ts)
```

```
## [1] 365
```

```
GaringerOzone.monthly.ts <- ts(GaringerOzone.monthly$Ozone.mean,  
                               start = c(2010, 1),  
                               end = c(2019, 12),  
                               frequency = 12)  
start(GaringerOzone.monthly.ts)
```

```
## [1] 2010 1
```

```
end(GaringerOzone.monthly.ts)
```

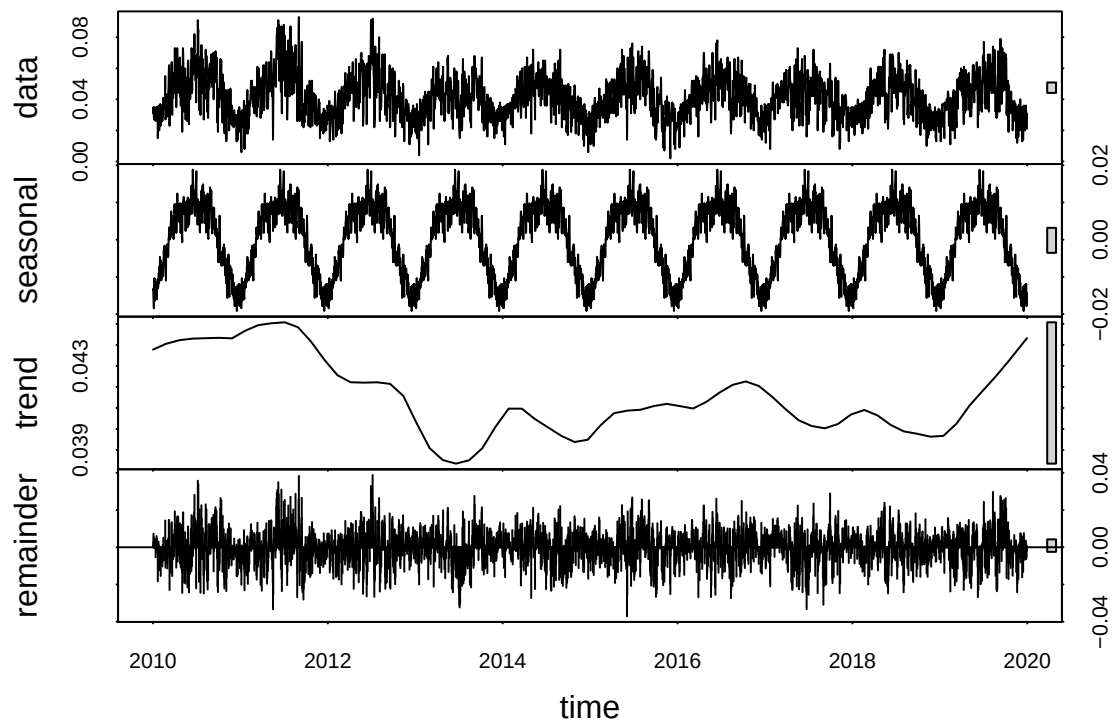
```
## [1] 2019 12
```

```
frequency(GaringerOzone.monthly.ts)
```

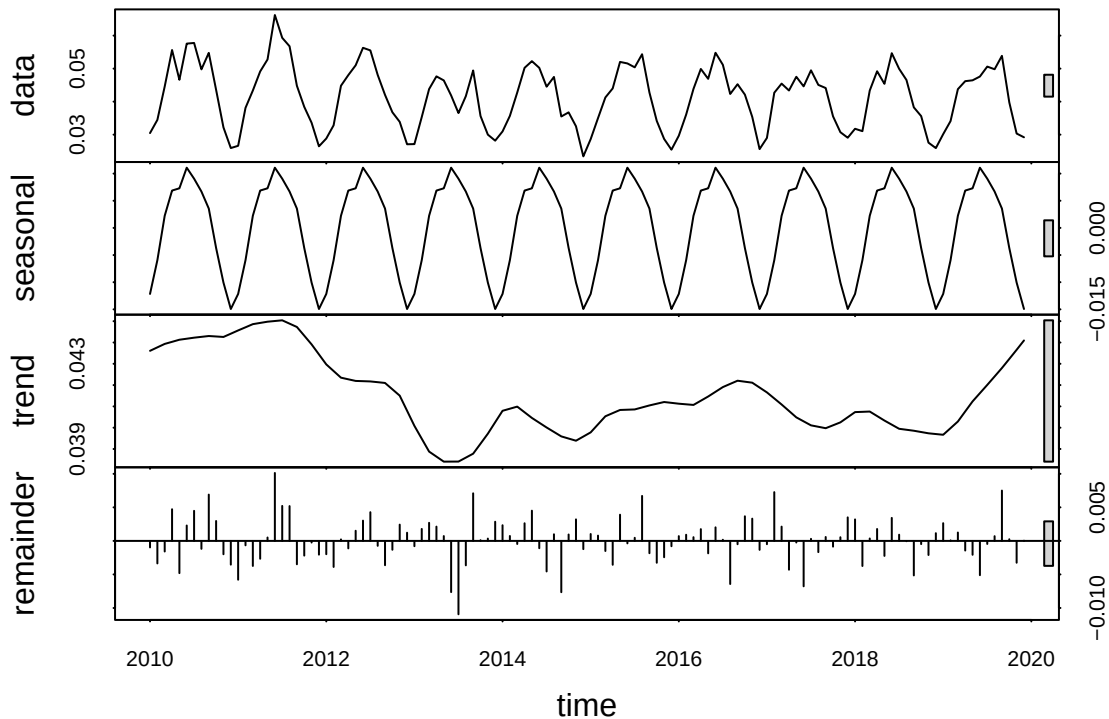
```
## [1] 12
```

11. Decompose the daily and the monthly time series objects and plot the components using the `plot()` function.

```
#11  
GaringerOzone.daily.decomp <- stl(GaringerOzone.daily.ts, s.window = "periodic")  
plot(GaringerOzone.daily.decomp)
```



```
GaringerOzone.monthly.decomp <- stl(GaringerOzone.monthly.ts, s.window = "periodic")
plot(GaringerOzone.monthly.decomp)
```



12. Run a monotonic trend analysis for the monthly Ozone series. In this case the seasonal Mann-Kendall is most appropriate; why is this?

```
#12
GaringerOzone.trend <- Kendall::SeasonalMannKendall(GaringerOzone.monthly.ts)

GaringerOzone.trend

## tau = -0.143, 2-sided pvalue =0.046724

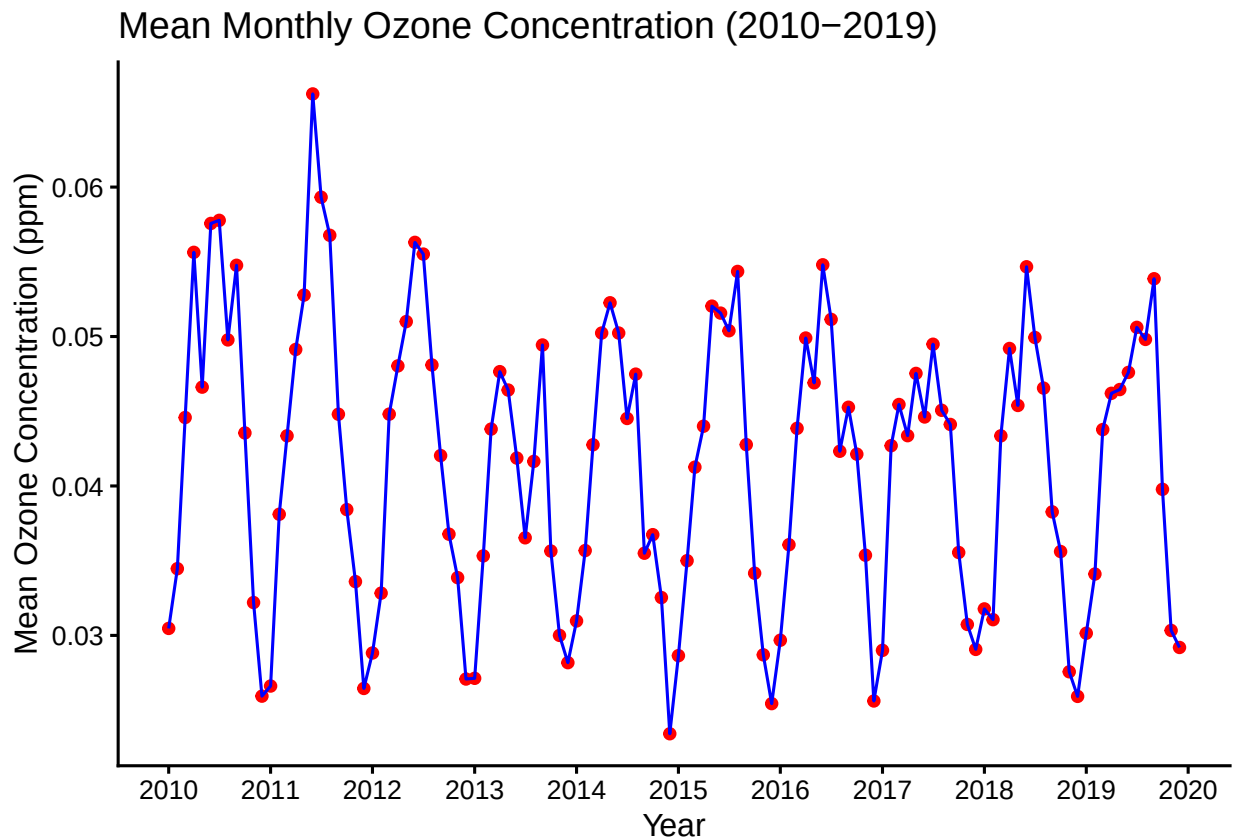
summary(GaringerOzone.trend)

## Score = -77 , Var(Score) = 1499
## denominator = 539.4972
## tau = -0.143, 2-sided pvalue =0.046724
```

Answer: Yes. It is most appropriate because the ozone time series exhibits strong and recurring seasonal patterns, as shown in the decomposition plots in question 11. The seasonal component demonstrates a consistent annual cycle, indicating that ozone concentrations vary systematically by month. A standard Mann-Kendall test would ignore this structure and violate the assumption of independence. In contrast, the seasonal Mann-Kendall test accounts for intra-annual variability by comparing observations within the same month across years. The test results ($\tau = -0.143$, 2-sided $p\text{-value} = 0.046724$) indicate a statistically significant but modest decreasing trend in ozone concentrations over the 2010s.

13. Create a plot depicting mean monthly ozone concentrations over time, with both a `geom_point` and a `geom_line` layer. Edit your axis labels accordingly.

```
# 13
ggplot(GaringerOzone.monthly, aes(x = Date, y = Ozone.mean)) +
  geom_point(color = "red") +
  geom_line(color = "blue") +
  labs(title = "Mean Monthly Ozone Concentration (2010–2019)",
       x = "Year",
       y = "Mean Ozone Concentration (ppm)") +
  scale_x_date(date_breaks = "1 year", date_labels = "%Y")
```



14. To accompany your graph, summarize your results in context of the research question. Include output from the statistical test in parentheses at the end of your sentence. Feel free to use multiple sentences in your interpretation.

Answer: Ozone concentrations at this station have slightly decreased over the 2010s, although the magnitude of change is small relative to strong and persistent seasonal variability, as shown by recurring annual peaks and troughs in the monthly time series. Both the smoothed trend and decomposition results support a modest downward trajectory over time. The seasonal Mann–Kendall test results ($\tau = -0.143$, -sided pvalue = 0.046724) indicate a statistically significant but modest decreasing trend in ozone concentrations over the 2010s.

15. Subtract the seasonal component from the `GaringerOzone.monthly.ts`. Hint: Look at how we extracted the series components for the `EnoDischarge` on the lesson Rmd file.

16. Run the Mann Kendall test on the non-seasonal Ozone monthly series. Compare the results with the ones obtained with the Seasonal Mann Kendall on the complete series.

```
#15
GaringerOzone.monthly.decomp <- decompose(GaringerOzone.monthly.ts)
seasonal_component <- GaringerOzone.monthly.decomp$seasonal
GaringerOzone.monthly.deseason <- GaringerOzone.monthly.ts - seasonal_component

#16

GaringerOzone.MK <- mk.test(GaringerOzone.monthly.deseason)
GaringerOzone.MK

##
## Mann-Kendall trend test
##
## data: GaringerOzone.monthly.deseason
## z = -2.6039, n = 120, p-value = 0.009216
## alternative hypothesis: true S is not equal to 0
## sample estimates:
##          S          varS          tau
## -1.149000e+03  1.943657e+05 -1.609356e-01
```

Answer: The Mann–Kendall test applied to the deseasonalized monthly ozone series indicates a statistically significant negative trend, with a stronger signal compared to the seasonal Mann–Kendall test on the original series. Specifically, the test shows a more negative tau value and a lower p-value (tau=-1.609356e-01 , p-value = 0.009216), suggesting that the decreasing trend becomes clearer after removing seasonal variability. In comparison, the seasonal Mann–Kendall test on the full series also detected a negative trend but with weaker significance (tau = -0.143, p-value = 0.046724)). This difference indicates that seasonal fluctuations can mask or dampen the underlying long-term trend, and removing seasonality allows for a more robust detection of the true monotonic change in ozone concentrations.